

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 26

Protect IdentityCenterAdministration with a Permission Boundary

IAM | IAM Identity Center | Permission Boundaries | Privilege Escalation Control

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Logged in to the **IAM** account using **AdministratorAccess**, created a customer-managed IAM policy named **SSOPermissionBoundary**, and attached it as a permissions boundary to the **IdentityCenterAdministration** IAM Identity Center permission set. The boundary allows expected delegated-administration activity while explicitly denying actions against the IdentityCenterAdministration permission set itself. The control was tested by signing in with the IdentityCenterAdministration role and validating that the role could not modify its own permission set.

BOUNDARY LOGIC

A permissions boundary does not grant permissions. It defines the maximum permission scope an identity can ever use. That distinction is essential because the boundary uses an Allow * statement to avoid breaking ordinary allowed work, then uses an explicit Deny against the protected permission set.

Business Problem

Project 25 gave IAM Identity Center administrators the ability to create and attach IAM policies so they could manage Identity Center permission sets. That creates a self-escalation risk: the same administrator could potentially attach a broader policy to their own permission set and grant themselves elevated access. This project uses a permissions boundary to preserve the administrative capability while blocking changes to the administrator's own permission set.

Objectives

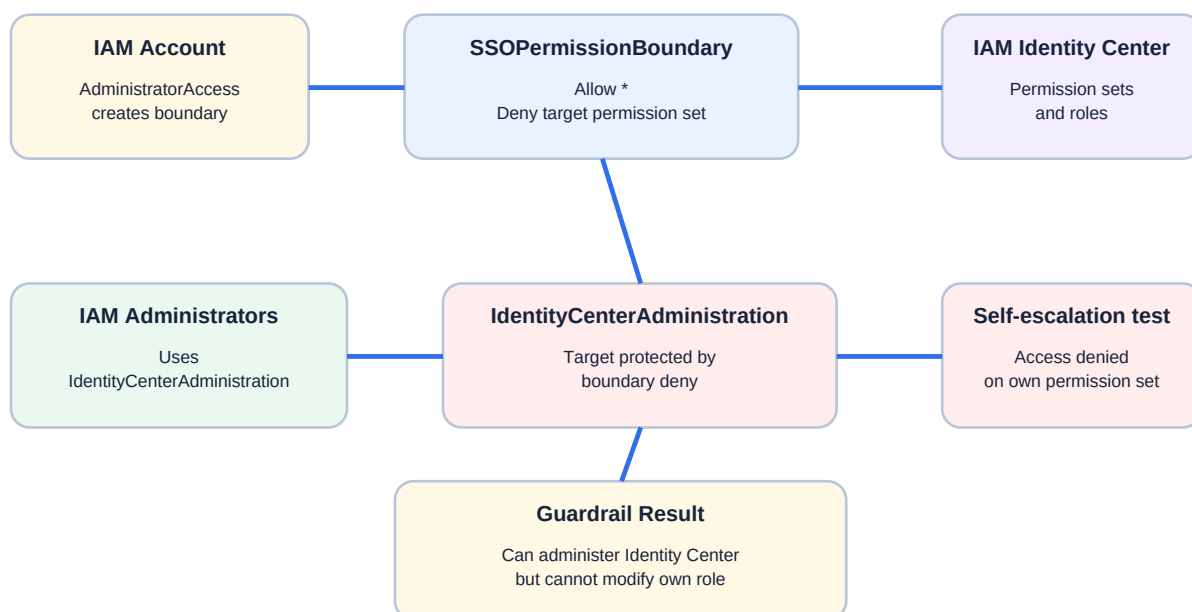
- Use IAM AdministratorAccess to create the boundary because IdentityCenterAdministration should not be able to create its own guardrail.
- Create a customer-managed policy named SSOPermissionBoundary.
- Use an AllowAllActions statement so normal allowed permissions are not blocked by the boundary.
- Use an explicit DenyActionsOnSpecificPermissionSet statement scoped to the IdentityCenterAdministration permission-set ARN.
- Attach the customer-managed policy as a permissions boundary to the IdentityCenterAdministration permission set.
- Wait for the permission set to reprovision to the IAM account.
- Test using the IdentityCenterAdministration role and verify access is denied on the protected permission set.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Services	IAM, IAM Identity Center
Account Context for Creation	IAM account with AdministratorAccess

Protected Permission Set	IdentityCenterAdministration
Boundary Policy	SSOPermissionBoundary
Boundary Policy Type	Customer-managed IAM policy
Boundary Logic	Allow all actions, then explicitly deny all actions on the protected permission set ARN
Attachment Location	IdentityCenterAdministration permission set - Permissions boundary section
Validation	IdentityCenterAdministration role sees access denied / needs permissions for protected permission set
Purpose	Prevent self-escalation while preserving delegated IAM Identity Center administration

Architecture



The boundary preserves delegated administration while preventing changes to the administrator's own permission set.

Figure 1 - Permission boundary preventing IdentityCenterAdministration from changing itself.

Implementation Summary

1. Signed into the IAM account using AdministratorAccess.
2. Collected the ARN of the IdentityCenterAdministration permission set.
3. Opened IAM Policies and created a new customer-managed policy.
4. Built a permissions-boundary policy with AllowAllActions and DenyActionsOnSpecificPermissionSet.

5. Replaced the placeholder resource value with the protected IdentityCenterAdministration permission set ARN.
6. Named the policy SSOPermissionBoundary and used a description explaining that it restricts IAM Identity Center administrators from escalating privileges.
7. Opened IAM Identity Center, selected Permission sets, and opened IdentityCenterAdministration.
8. Set the customer-managed policy SSOPermissionBoundary as the permissions boundary for IdentityCenterAdministration.
9. Waited for the permission set to reprovision to the IAM account.
10. Signed out and tested with the IdentityCenterAdministration role.
11. Confirmed the role was denied permission to modify or access its own protected permission set.

Boundary Policy Pattern

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowAllActions",
      "Effect": "Allow",
      "Action": "*",
      "Resource": "*"
    },
    {
      "Sid": "DenyActionsOnSpecificPermissionSet",
      "Effect": "Deny",
      "Action": "*",
      "Resource": "<IdentityCenterAdministration permission-set ARN>"
    }
  ]
}
```

Control Map

Allow statement	AllowAllActions	Prevents the boundary from blocking every action by default.
Deny statement	DenyActionsOnSpecificPermissionSet	Blocks actions against the protected IdentityCenterAdministration permission set.
Boundary target	IdentityCenterAdministration	Protects the role used by IAM Administrators.
Boundary policy	SSOPermissionBoundary	Reusable managed policy used as the permissions boundary.
Test role	IdentityCenterAdministration	Validates that the administrator cannot modify its own permission set.

Evidence

Figure 2 - Permission Boundary JSON Pattern

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowAllActions",
      "Effect": "Allow",
      "Action": "*",
      "Resource": "*"
    },
    {
      "Sid": "DenyActionsOnSpecificPermissionSet",
      "Effect": "Deny",
      "Action": "*",
      "Resource": "INSERT YOUR ARN HERE"
    }
  ]
}
```

The policy JSON evidence shows an AllowAllActions statement for Action * and Resource *, followed by an explicit DenyActionsOnSpecificPermissionSet statement that denies Action * against the targeted permission-set ARN.

Figure 3 - SSOPermissionBoundary Policy Review

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name

Enter a meaningful name to identify this policy.

SSOPermissionBoundary

Maximum 128 characters. Use alphanumeric and '+', '@', '-' characters.

Description - optional

Add a short explanation for this policy.

Restrict an IAM Identity Center admin fro escalating their privileges

Maximum 1,000 characters. Use alphanumeric and '+', '@', '-' characters.

This policy defines some actions, resources, or conditions that do not provide permissions. To grant access, policies must have an action that has an applicable resource or condition. For details, choose Show remaining.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Explicit deny (1 of 472 services)

Service	Access level	Resource	Request condition
IAM Identity Center	Full: Tagging Limited: List, Permissions management, Read, Write		None

The policy review evidence shows the customer-managed policy name SSOPermissionBoundary and a description stating that it restricts an IAM Identity Center administrator from escalating privileges. The policy summary shows an explicit deny for IAM Identity Center.

Figure 4 - IdentityCenterAdministration Test Shows Access Denied

Permission sets (4)

Permission sets define the level of access that users in AWS IAM Identity Center have to their assigned AWS accounts. The names of permission sets appear as available roles in the AWS access portal. Users who are assigned to multiple AWS permission sets can sign in to the AWS access portal, choose an account, and then choose a role that AWS created from an assigned permission set. [Learn more](#)

Q Search permission sets by name, ARN, or ID (i.e., ps-abcdefg123456789)

Permission set	Description	ARN	Provisioning status	Creation time
☐	-	arn:aws:sso::permissionSet/ssoins-79073aee4...	You need permissions	less than a minute ago
☐ ReadOnlyAccess	-	arn:aws:sso::permissionSet/ssoins-79073aee4...	Provisioned	5 days ago
☐ AdministratorAccess	Full admin	arn:aws:sso::permissionSet/ssoins-79073aee4...	Provisioned	7 days ago
☐ SecurityFullAdmin	-	arn:aws:sso::permissionSet/ssoins-79073aee4...	Provisioned	5 days ago

The validation evidence shows the IAM Identity Center Permission sets page with the protected permission set showing You need permissions while other permission sets remain visible and provisioned. This demonstrates the boundary preventing the IdentityCenterAdministration role from accessing or modifying its own permission set.

Security Analysis

The design prevents a delegated IAM Identity Center administrator from changing the permission set that grants their own access. This directly reduces self-escalation risk.

The AllowAllActions statement is required because a permissions boundary is a maximum-permissions guardrail. Without an allow path in the boundary, default deny would block all effective access.

The explicit Deny against the IdentityCenterAdministration permission set takes priority over any allow from the attached managed policies. This is the key enforcement mechanism.

The boundary does not fully solve every Identity Center escalation path. A remaining risk is that a privileged identity administrator may be able to alter group membership, depending on the identity store and federation design.

This is a more advanced IAM pattern than simply assigning or removing managed policies because it controls the maximum effective power of an already privileged identity.

Lessons Learned

Permissions boundaries limit what an IAM principal can do; they do not grant new permissions.

Boundaries are especially useful when an administrator needs to manage IAM or permission sets, but should not be able to use that power to escalate their own role.

An explicit Deny scoped to a sensitive permission-set ARN can protect a role from self-modification.

Testing with the restricted role is required. Merely creating and attaching the boundary does not prove the control works.

Production Improvements

- Monitor changes to permission boundaries and permission-set policy attachments with CloudTrail and EventBridge.
- Protect the SSOPermissionBoundary policy itself from modification by the same administrator persona.
- Use an external identity provider such as Entra ID or Okta to reduce Identity Center group self-assignment risk.
- Create alerts for membership changes to IAM Administrators and Administrators groups.
- Document the break-glass path for temporarily modifying the protected permission set.
- Review whether the Deny should include the boundary policy ARN itself in a production environment.

Skills Demonstrated

Permissions boundaries	Explicit Deny	IAM Identity Center
Self-escalation prevention	Permission-set hardening	Delegated admin controls
IAM managed policies	Access testing	Least privilege

Resume Bullet

RESUME

Implemented an AWS IAM permissions boundary for IAM Identity Center administration by creating SSOPermissionBoundary, attaching it to IdentityCenterAdministration, and validating that delegated admins cannot modify their own permission set.

STAR Interview Story

Situation: IAM Identity Center administrators had permission to create and attach IAM policies, which created a risk that they could modify their own permission set and escalate privileges.

Task: Add a guardrail that preserves delegated administration while preventing self-modification of the administrator permission set.

Action: I created the SSOPermissionBoundary customer-managed policy, used AllowAll plus an explicit Deny against the IdentityCenterAdministration permission-set ARN, attached it as a permissions boundary, and tested with the restricted role.

Result: The IdentityCenterAdministration role was denied access to its own protected permission set, reducing self-escalation risk while preserving the delegated administration model.

Version History

v1.0.0	July 16, 2026	Initial documented implementation. Created SSOPermissionBoundary, attached it as a permissions boundary to IdentityCenterAdministration, tested access denial using the IdentityCenterAdministration role, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Permissions Boundaries Made Easy.
- User-provided SSOPermissionBoundary JSON screenshot.
- User-provided SSOPermissionBoundary policy review screenshot.
- User-provided IdentityCenterAdministration access-denied validation screenshot.
- Project 25 - IAM Identity Center Manage Policies Permission Set.